



Comparative Evaluation of Satellite Image Segmentation Methods for Coastline Extraction: Spectral Indices vs. Deep Learning

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Area 4: Pattern Recognition and Computer Vision in Applied Sciences – Geographic Information Systems, satellite images and geopositioning

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1. Introduction

Motivation and context



Coastal regions are among the **most dynamic**, **densely populated**, and **socio-economically vital areas on Earth**.



However, these environments are under **constant threat** from both **natural** processes and **anthropogenic pressures**, **exacerbated by climate change**.

Coastal monitoring

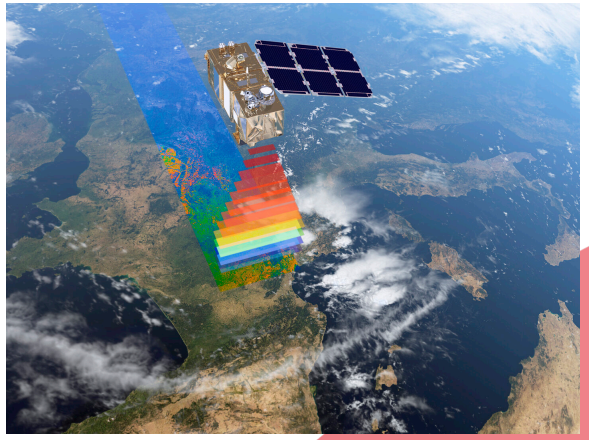
Field campaigns



Video systems



Optical satellites

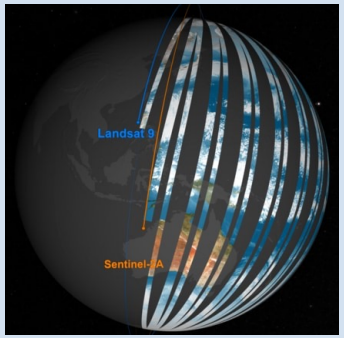


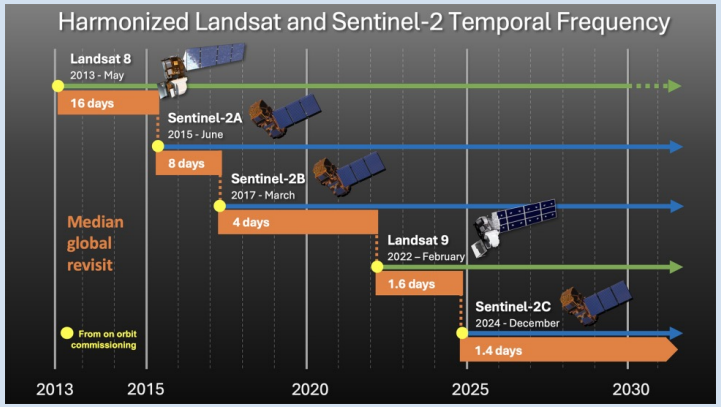
Motivation and context



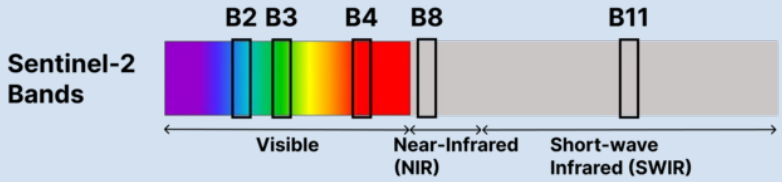
Key points of optical satellite for coastal monitoring (Landsat 4-9 and Sentinel-2)

Global coverage and high revisit frequency





Hyperspectral data



Free open data and easily access (e.g. GEE API)

Motivation and context

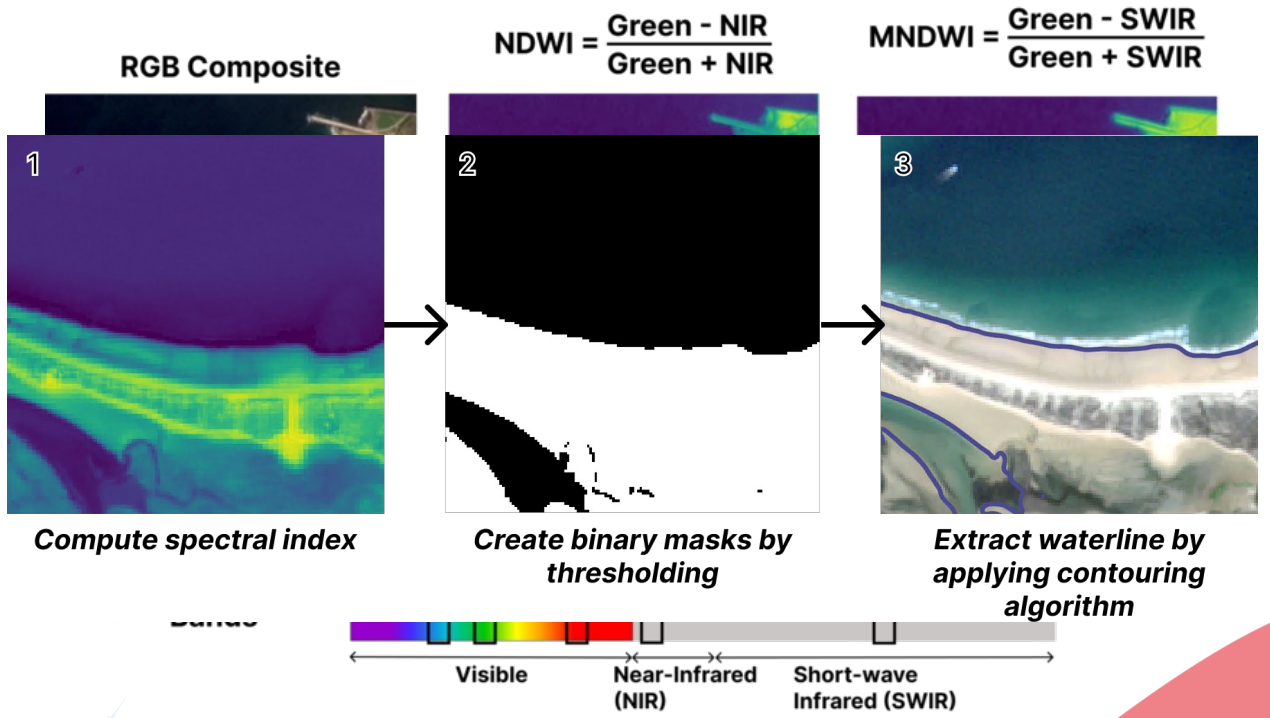


What products can be derived from satellite imagery and how

Satellite-derived Waterline (SDW)

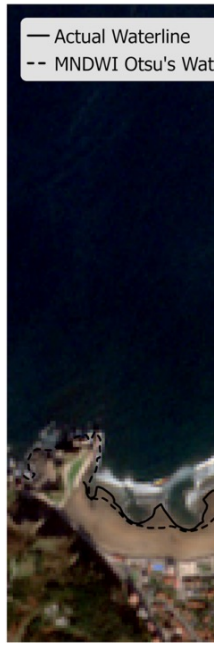


Traditional SDW extraction via spectral indexes + thresholding

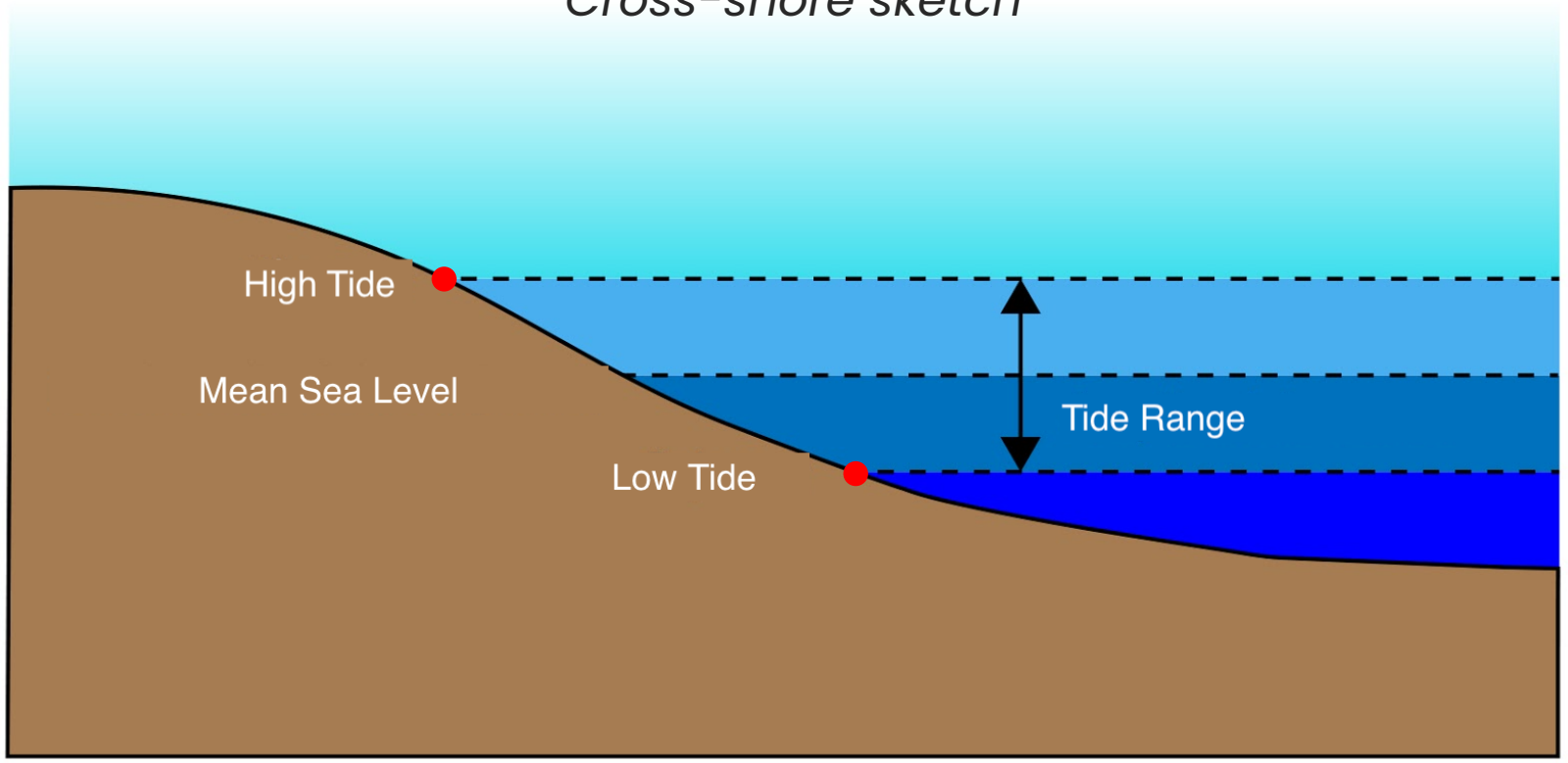


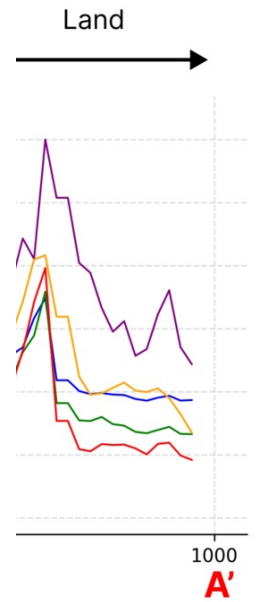
Motivation and context

Challenges and limitations in macrotidal environments



Cross-shore sketch

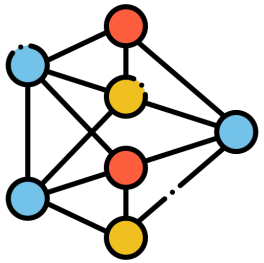




Motivation and context



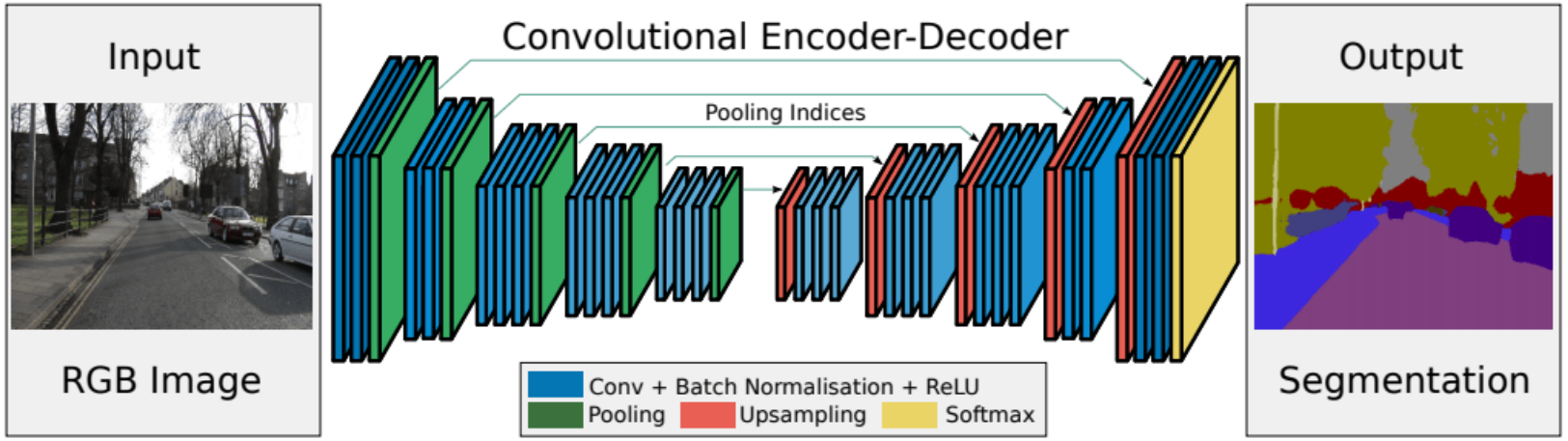
Are there any more robust alternatives?



Semantic Segmentation models (Deep Learning)

→

Capable of understanding spatial context (do not rely on pixel-by-pixel calculations)



Work objectives



Benchmarking between traditional index-based vs. Deep Learning-based models



To generate Ground Truth photo-interpreted datasets of both coastal masks and waterlines

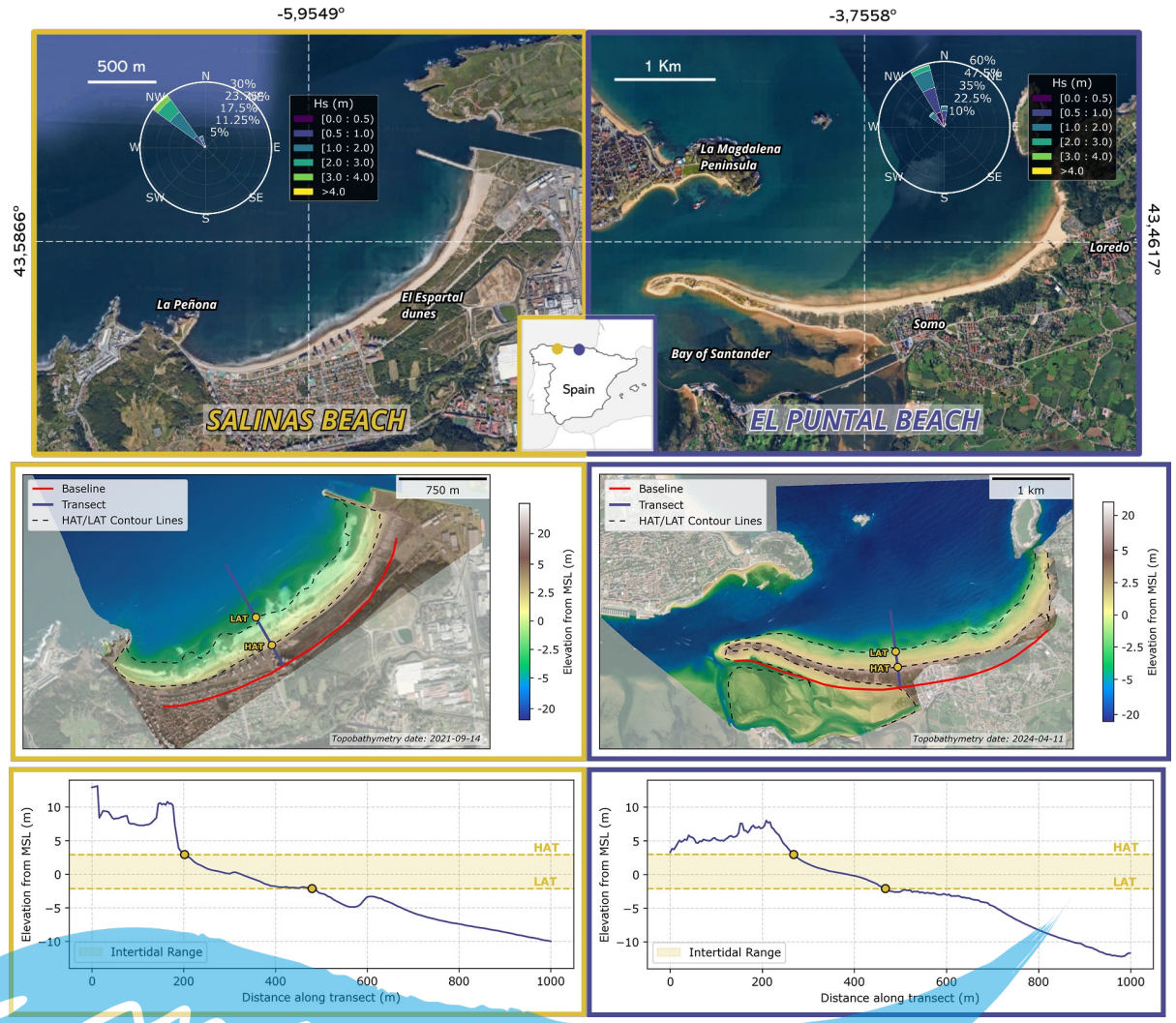
To evaluate and compare the performance degradation under adverse conditions (oceanographic, morphological, etc.)



2. Materials and Methods

Materials and Methods

Selection of study areas



Common features

- Both beaches have microtidal regimes (>4m)

Key differences

- Wave exposition
- Sand color and darkness

Materials and Methods

Ground-Truth (GT) generation



Sentinel-2 downloading images

Waterline photointerp.

Waterlines (.SHP): 144 Salinas and 141 El Puntal

As reference to evaluate both index-based models vs. DL-based models

Pixel class annotation *(water, white water, wet sand and dry sand)*

To retrain the DL models

Multiclass masks (.TIF)

Materials and Methods



Selection of models

SPECTRAL INDEXES

4 indexes:
- NDWI
- MNDWI
- AWEInsh
- SCoWI

3 thresholdings:
- Zero
- Otsu
- LM Otsu

4 x 3 = 12 models

SCoWI (LM Otsu)*

DEEP LEARNING

3 architectures:
- U-Net (2015)
- DeepLabV3+ (2018)
- SegFormer (2021)

4 encoders:
- EfficientNet (B4 y B7)
- MiT (B2 y B4)

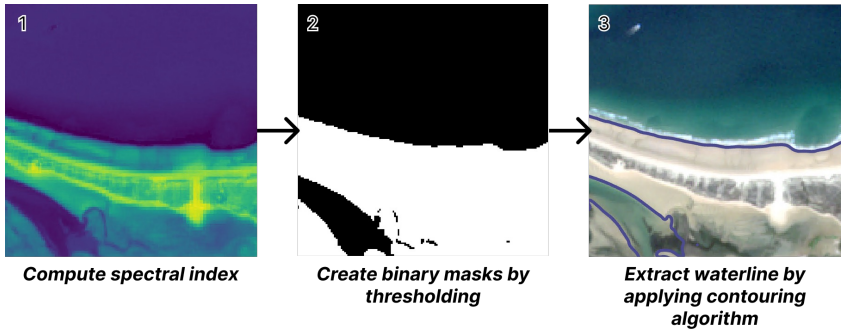
3 x 4 = 12 models

*Bergsma, E. W., Klotz, A. N., Artigues, S., Graffin, M., Prenowitz, A., Delvit, J. M., & Almar, R. (2024). Shoreliner: A sub-pixel coastal waterline extraction pipeline for multi-spectral satellite optical imagery. Remote Sensing, 16(15), 2795.

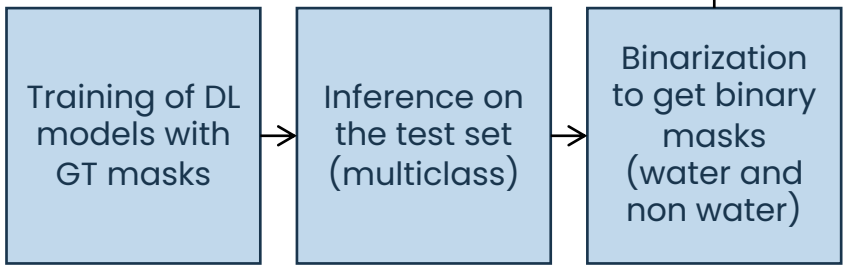
Materials and Methods

Waterlines to evaluate

Pipeline for extracting waterlines for index-based models



Pipeline for extracting waterlines for DL-based models

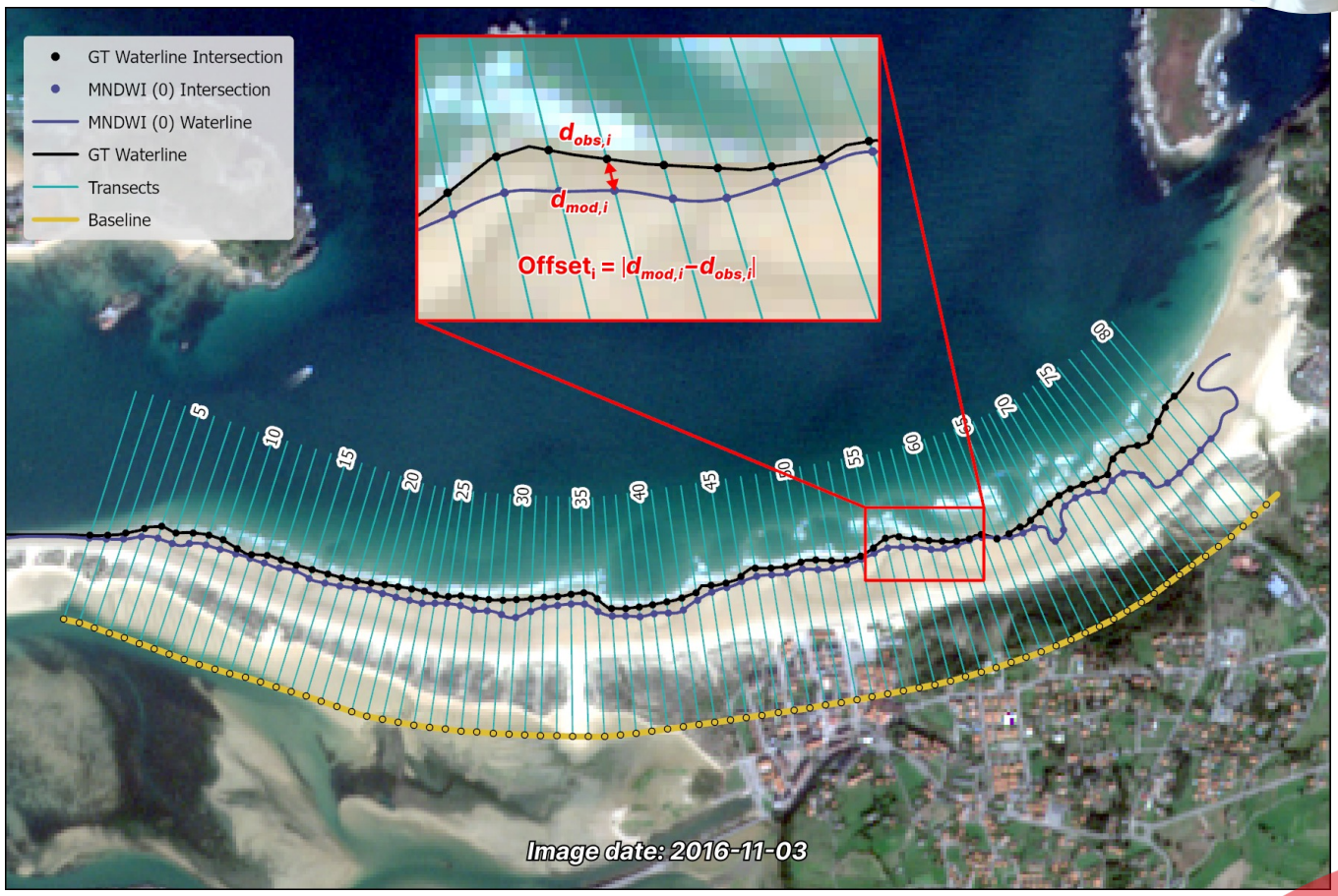


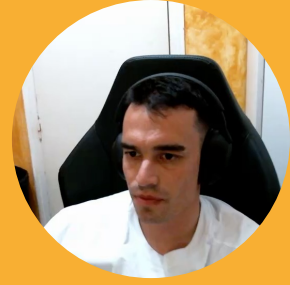
Patches, train test split, data augmentation, etc.

Materials and Methods



Cross-shore distances and error (offset) calculation

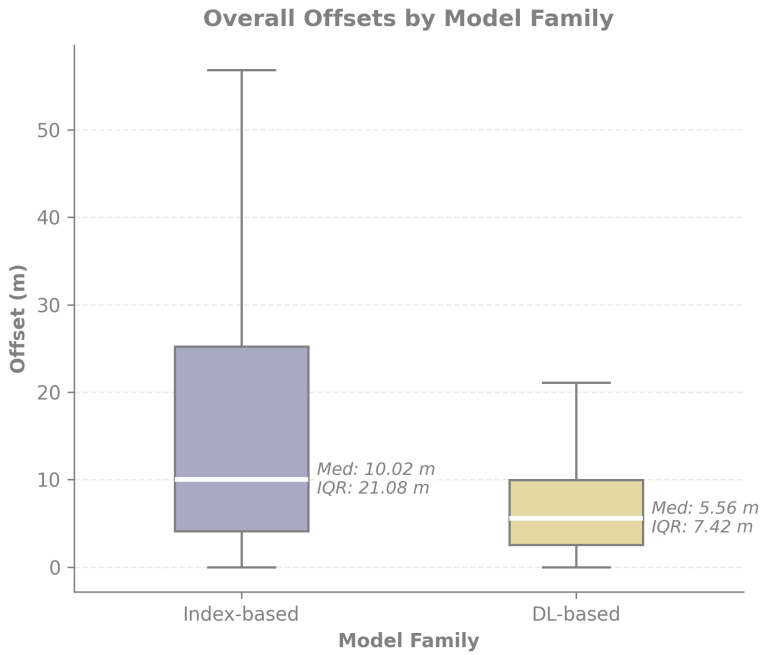


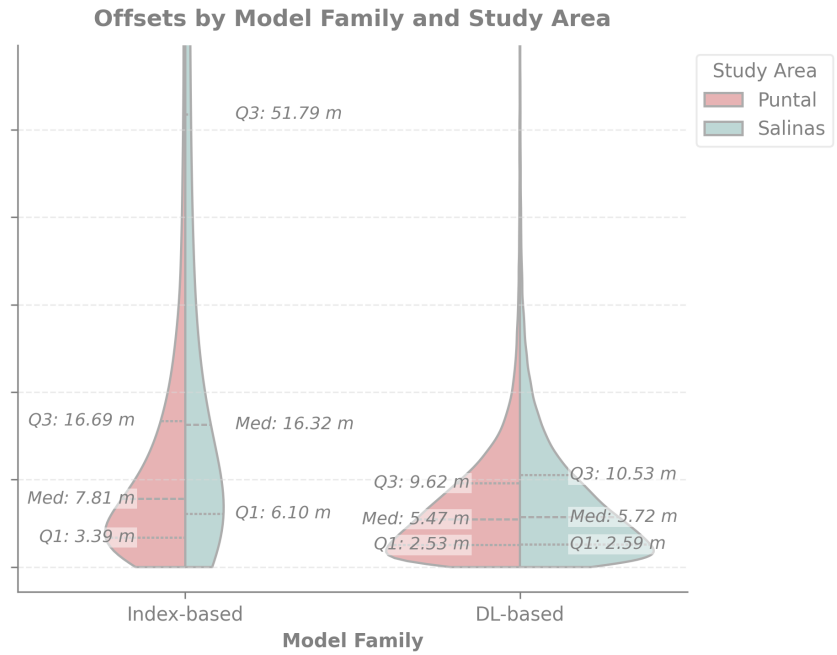


3. Results

Results

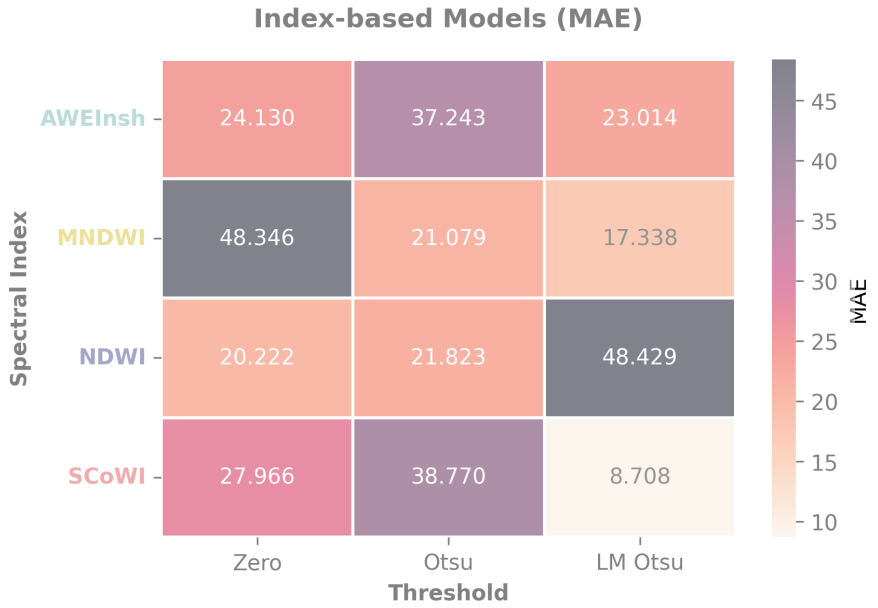
Overarching Family Performance (index-based and DL-based)

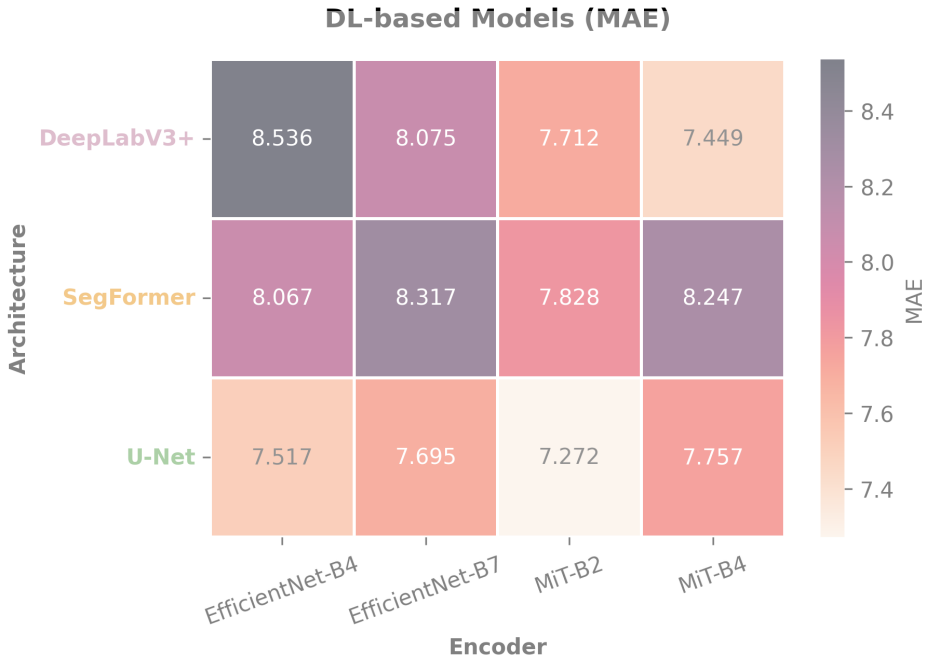




Results

Intra-Family Model Assessment and Spatial Bias

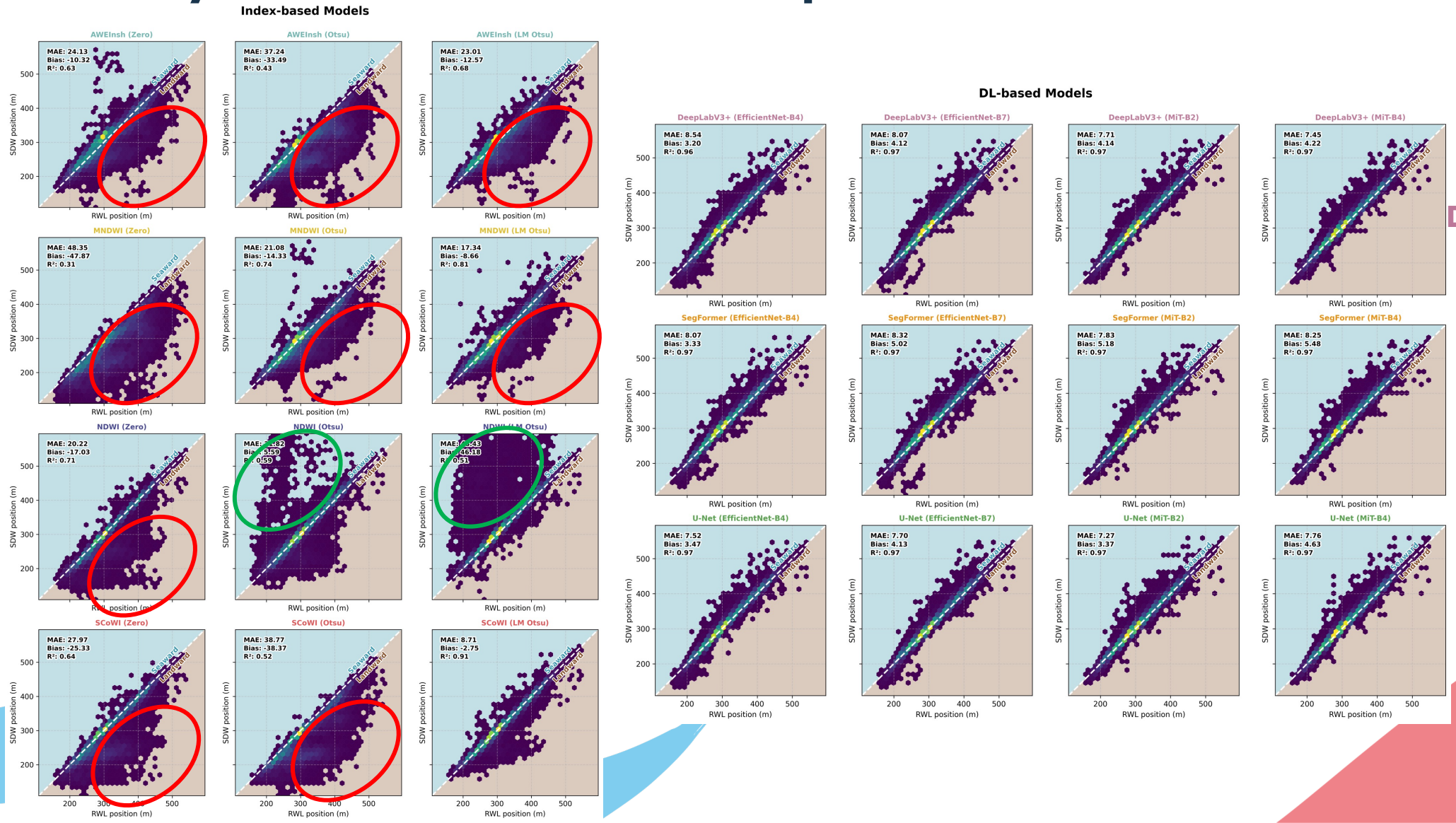




Results

Intra-Family Model Assessment and Spatial Bias





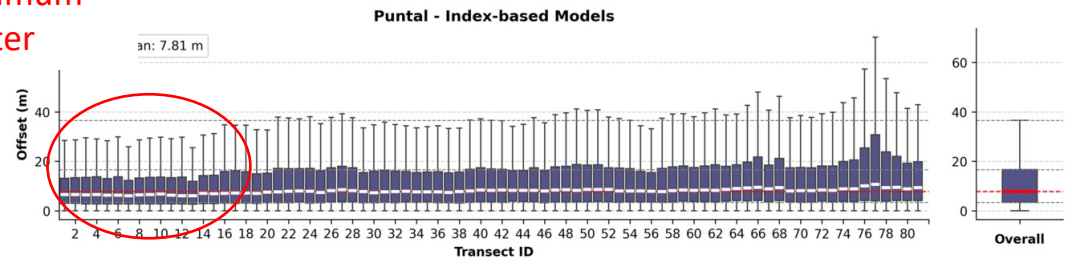
Results

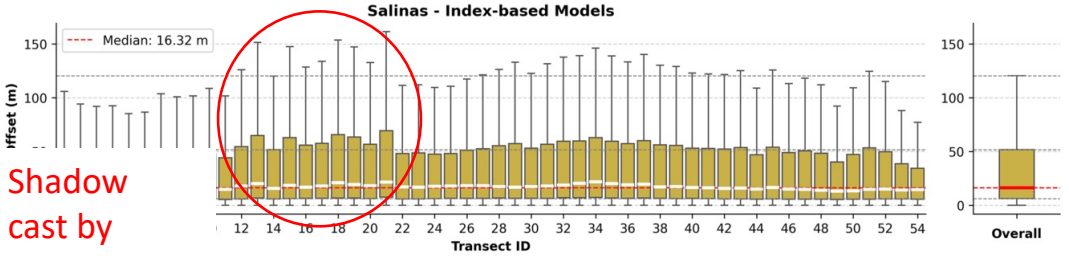
Analysis of sources of error (spatial and hydrodynamic)





Maximum shelter

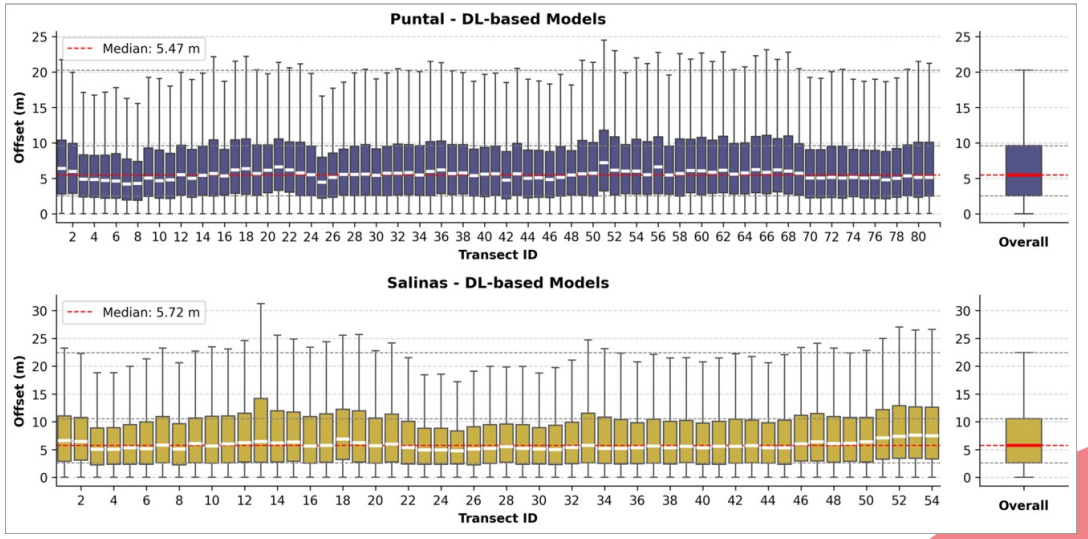




Shadow cast by skyscrapers

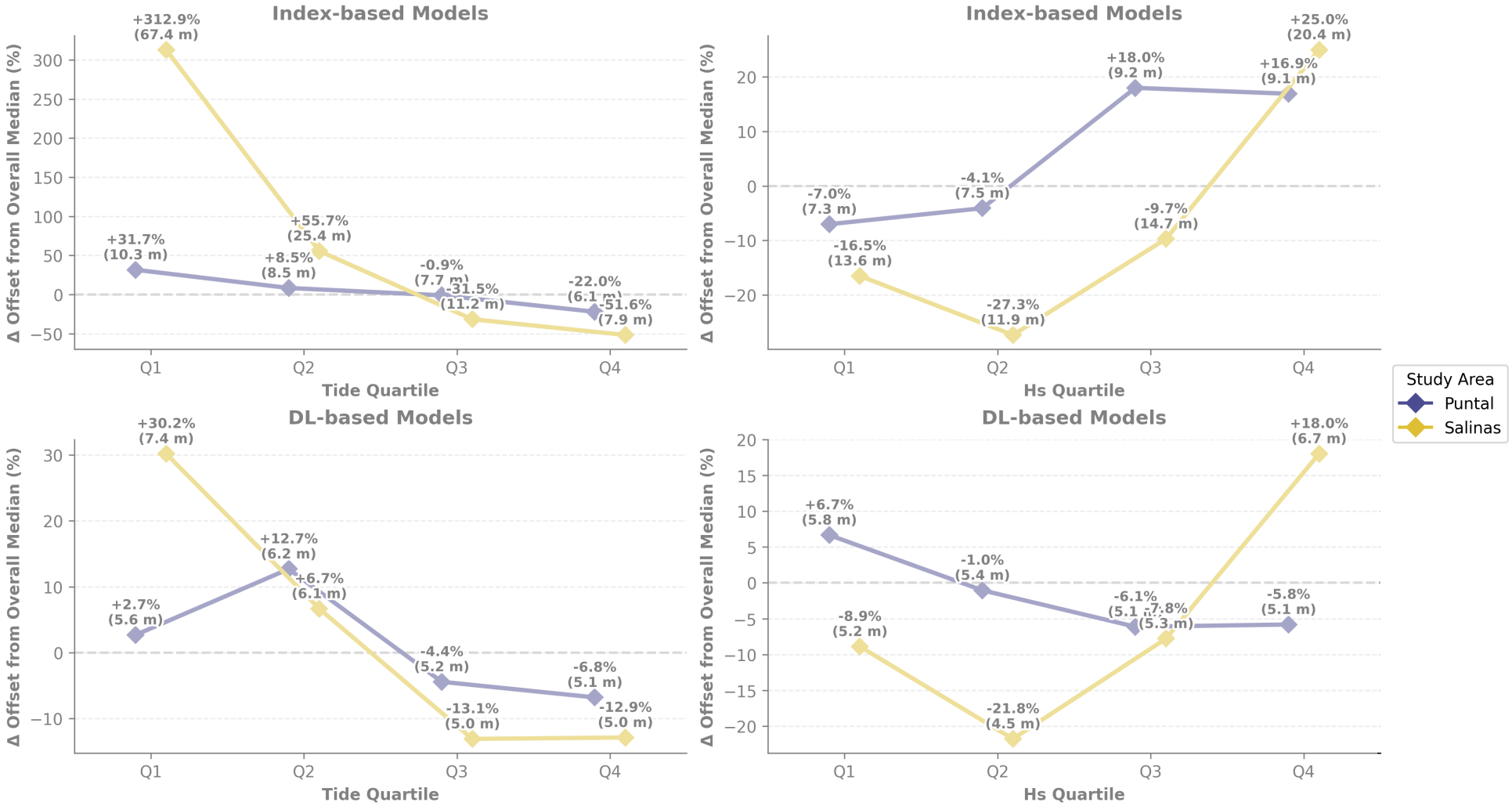






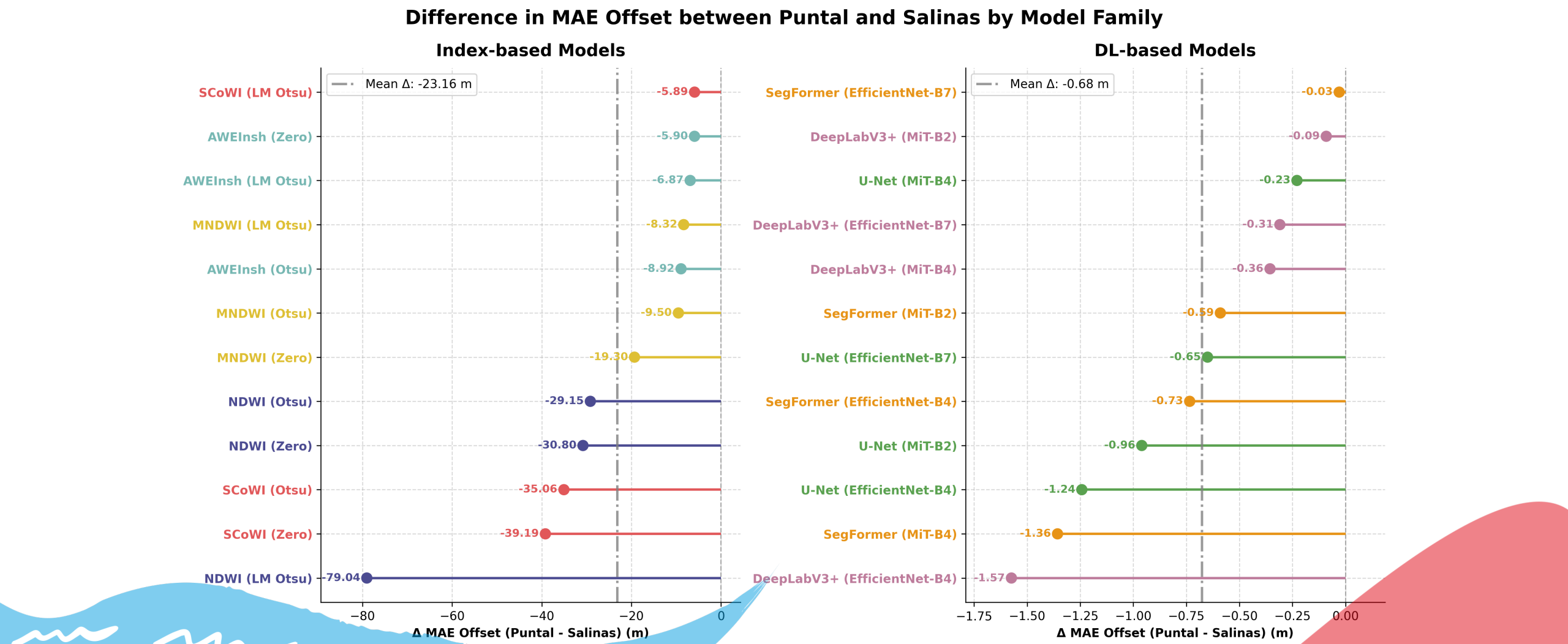
Results

Analysis of sources of error (spatial and hydrodynamic)



Results

Cross-Site Generalization and Robustness

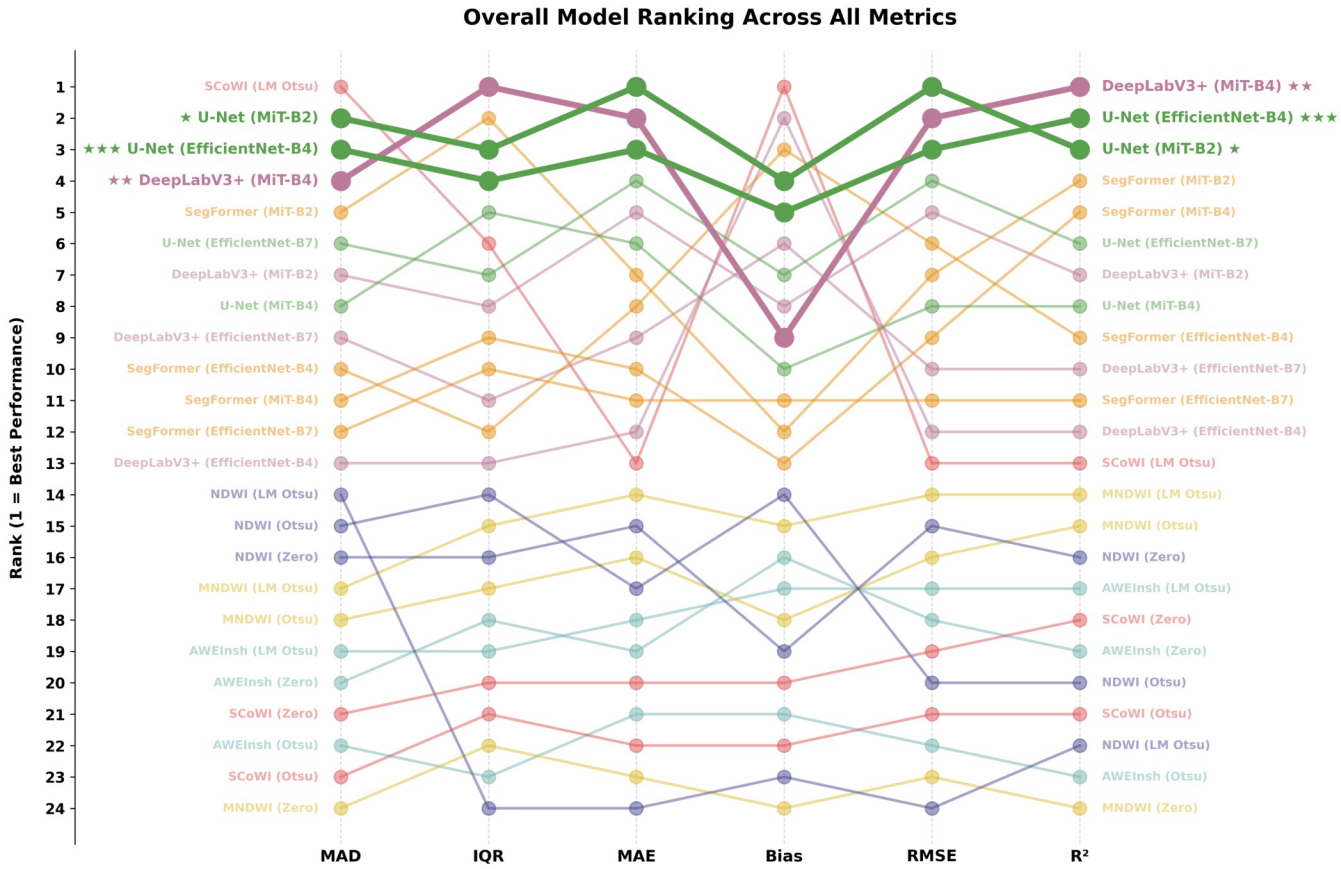


Results

Final Multi-Criteria Ranking

Overall - Model Metrics

	MAD	IQR	MAE	Bias	RMSE	R^2
AWEInsh (Zero)	15.12	33.90	24.13	-10.32	38.40	0.63
AWEInsh (LM Otsu)	13.88	34.48	23.01	-12.57	36.00	0.68
AWEInsh (Otsu)	19.90	54.36	37.24	-33.49	57.03	0.43
MNDWI (Zero)	31.06	53.18	48.35	-47.87	71.27	0.31
MNDWI (LM Otsu)	10.65	23.57	17.34	-8.66	27.08	0.81
MNDWI (Otsu)	11.76	29.56	21.08	-14.33	33.99	0.74
NDWI (Zero)	10.07	25.13	20.22	-17.03	32.36	0.71
NDWI (LM Otsu)	7.28	74.90	48.43	46.18	93.06	0.51
NDWI (Otsu)	8.36	15.82	21.82	5.59	45.49	0.59
SCoWI (Zero)	18.07	38.89	27.97	-25.33	40.63	0.64
SCoWI (LM Otsu)	**4.90**	9.96	8.71	**-2.75**	16.67	0.91
SCoWI (Otsu)	25.31	47.34	38.77	-38.37	55.62	0.52
DeepLabV3+ (EfficientNet-B4)	6.13	11.95	8.54	3.20	12.62	0.96
DeepLabV3+ (EfficientNet-B7)	5.70	10.66	8.07	4.12	12.33	0.97
DeepLabV3+ (MiT-B2)	5.53	10.16	7.71	4.14	11.81	0.97
DeepLabV3+ (MiT-B4)	5.36	**9.50**	7.45	4.22	11.49	**0.97**
SegFormer (EfficientNet-B4)	5.88	10.96	8.07	3.33	11.96	0.97
SegFormer (EfficientNet-B7)	6.08	10.32	8.32	5.02	12.47	0.97
SegFormer (MiT-B2)	5.44	9.77	7.83	5.18	12.13	0.97
SegFormer (MiT-B4)	6.01	10.31	8.25	5.48	12.32	0.97
U-Net (EfficientNet-B4)	5.26	9.84	7.52	3.47	11.53	0.97
U-Net (EfficientNet-B7)	5.46	10.14	7.70	4.13	11.75	0.97
U-Net (MiT-B2)	5.11	9.81	**7.27**	3.37	**11.35**	0.97
U-Net (MiT-B4)	5.55	9.85	7.76	4.63	12.27	0.97
	MAD	IQR	MAE	Bias	RMSE	R^2





4. Conclusions and Future Work

Conclusions



On average, deep learning architectures **halve the error** compared with spectral index models.

Deep Learning architectures provide a **robust**, **environmentally resilient**, and **geographically invariant** solution.

The **U-Net** architecture has yielded the best results.

Future Work



- Global Expansion of the Training Dataset
- Sub-pixel Resolution Enhancement
- Development of an Open-Source Operational Tool



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